WORKING PAPER

The Rural Mobilization Gap in a U.S. Place-Based Tax Credit

Intermediary Selection versus Market Structure in the New Markets Tax Credit

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Abstract

The U.S. New Markets Tax Credit (NMTC), enacted in 2000, gives private investors a 39% federal tax credit for routing capital through certified Community Development Entities (CDEs) into projects in low-income census tracts. Over FY2001–FY2022 the program channeled \$66.6 billion of subsidized investment into 8,024 projects worth \$120.9 billion, and each dollar of subsidized investment sat alongside \$0.82 of other capital. Federal cost is the 39% credit on the equity basis, so the ratio against tax expenditure is roughly \$2.09. Non-metropolitan tracts received 19.6% of dollars, close to the program's 20% administrative target, yet project-level leverage was lower there (median 1.07× against 1.19× metro).

I decompose this rural mobilization penalty with project-level fixed-effects regressions on the CDFI Fund's public release. The raw gap is a precise −0.262, and it does not survive the introduction of intermediary fixed effects: the within-CDE estimate is −0.047, the median-regression analog is −0.001, and both the extensive and intensive margins are null within intermediary. An order-invariant Gelbach decomposition assigns −0.185 of the raw gap, 86% of the explained movement, to CDE identity, and adding the release's purpose-of-investment field as a fourth block, which no prior specification used, leaves 88.1% of that movement with

CDE identity. The rural leverage gap lies between intermediaries and not within them.

One exception survives scrutiny. Across twenty-four subgroups, rural commercial real-estate rehabilitation carries a within-CDE coefficient of −0.439 (CDE-clustered SE 0.102), the only cell to survive a multiple-comparison correction. It withstands every outcome transformation, leave-one-intermediary and leave-one-state drop, randomization inference, a wild cluster bootstrap, and placebo tests on adjacent purposes. Because the purpose dimension was added after an earlier draft flagged real estate, I report it as exploratory. A cross-CDE excess-mass test finds no bunching at the 20% non-metropolitan target, in deal counts or dollars, over the full period or after the proportionality rule entered the code in 2006. Compliance may operate through allocation agreements, which the deployment data do not record. For policy, the relevant levers concern intermediary selection and capacity.

JEL H25 · H81 · R51 · G28

Keywords New Markets Tax Credit · place-based policy · blended finance · mobilization ratio · Community Development Entities · rural finance

1 Introduction

The blended-finance literature has long debated which structures best mobilize private capital alongside public concessional finance [Convergence Finance, 2024, Attridge and Engen, 2019, Lankes, 2021]. Its recurring empirical problem is measurement. The "mobilization ratio," private dollars moved per public dollar committed, is rarely observed at the project level. Most evidence comes from survey reports or coarse program aggregates.

The U.S. New Markets Tax Credit offers a setting where the financing structure behind that ratio becomes visible at the deal level. For every project, the CDFI Fund's public release records both the subsidized investment that the intermediary places (the Qualified Low-Income Community Investment, or QLICI) and total project cost. I use their ratio, the amount of project capital accompanying each dollar of subsidized investment. Section 3 defines the measure more fully. The government's cost is the credit claimed against the equity basis, so fiscal leverage uses a different denominator. I report both denominators when magnitudes matter. The panel follows one instrument in one country for twenty-two years and roughly 8,000 projects, with the financing structure observed directly.

Projects in non-metropolitan tracts carry less capital beyond the subsidized investment than projects in metropolitan ones. The question is where that gap originates. It may reflect *market structure*, with rural projects fundamentally less leverageable. It may instead reflect *intermediary selection*: the Community Development Entities active in rural tracts may mobilize less capital everywhere they operate. The policy prescriptions differ. Market structure points toward redesigning the credit; selection points toward reallocating it.

The estimates point to composition. The raw rural penalty in project-level leverage is −0.262 and precisely estimated. Origination-year, project-type, and state fixed effects move it to −0.172. CDE fixed effects compare rural and urban deals *made by the same intermediary*, using the 163 of 343 CDEs that work on both sides of the line. With those effects, the point estimate falls to −0.047; a fully nested model retaining state effects gives −0.060. The mean regression remains imprecise (cluster-robust 95% CI [−0.245, +0.152]) and can reject only within-CDE penalties larger than 0.21. The other estimates are tighter. The median within-CDE penalty is −0.001 (SE 0.008). Within a CDE, rural deals are equally likely to draw capital beyond the subsidized investment and draw no less when they do. The spatial difference comes from which intermediaries originate the deals.

The NMTC evaluation literature has examined program targeting and neighborhood outcomes extensively [Gurley-Calvez et al., 2009, Freedman, 2012, Harger and Ross, 2016, Freedman and Kuhns, 2018, Theodos et al., 2020, 2022, Corinth et al., 2024], with program-evaluation foundations in Abravanel et al. [2013]; the place-based policy literature supplies the welfare frame [Glaeser and Gottlieb, 2008, Kline and Moretti, 2014, Neumark and Simpson, 2015, Diamond and McQuade, 2019]. Mobilization has received much less attention in either literature. I have found no prior econometric study that treats project-level financing structure as the main NMTC outcome. Doing so brings the mobilization framework to work otherwise centered on neighborhood employment, demographic change, and price effects.

I then use CDE fixed effects to separate the within-CDE rural penalty from the between-CDE selection component. Descriptive work by Theodos et al. [2021] motivates this decomposition without estimating it. I also apply the excess-mass logic of Chetty et al. [2011] and Kleven [2016] to the cross-CDE distribution of non-metro deployment shares around the program's 20% target. I have not found this test in the published literature. There is no excess mass at the 20% line in deal counts or dollars, either over the full period or after the proportionality rule entered the code. One explanation is that compliance operates through allocation agreements absent from the public release.

Section 2 describes the program and its CDE intermediary layer. Section 3 covers the public release and construction of the leverage outcome. Section 4 sets out the fixed-effects design and its limits, and Section 5 reports the decomposition, heterogeneity, robustness, and bunching diagnostic. Section 6 gives the policy reading; Section 7 concludes. A causal companion design, using a regression discontinuity at the low-income-community eligibility threshold, requires a tract-level ACS merge and remains for follow-on work.

2 Institutional background

2.1 Program mechanics

The NMTC, enacted in the Community Renewal Tax Relief Act of 2000, awards investors a 39% federal tax credit, claimed over seven years, on Qualified Equity Investments (QEIs) made into certified Community Development Entities. CDEs in turn deploy that capital as Qualified Low-Income Community Investments into Qualified Active Low-Income Community Businesses (QALICBs) located in eligible census tracts. The public release distinguishes four QALICB types: real-estate projects (RE), non-real-estate operating businesses (NRE), special-purpose entities (SPE), and CDE-to-CDE loans. These mechanics put the CDE at the center of the program. It holds the allocation and originates the deals, then structures the capital stack around the subsidized tranche. Figure 1 shows the sequence. Capital moves from the investor through the CDE to the project, while the tax credit returns to the investor.

2.2 Eligibility and the 20% non-metro minimum

Eligible low-income community (LIC) tracts are defined by a poverty rate of at least 20% or median family income at or below 80% of the area benchmark, with a targeted-populations override. Relevant here, a later amendment directs the Treasury to write rules ensuring that non-metropolitan counties receive a proportional allocation of qualified equity investments. That instruction is IRC §45D(i)(6), added by the Tax Relief and Health Care Act of 2006 (P.L. 109–432, §102(b)), so it postdates the program's creation in the Community Renewal Tax Relief Act of 2000 (P.L. 106–554, §121), leav-

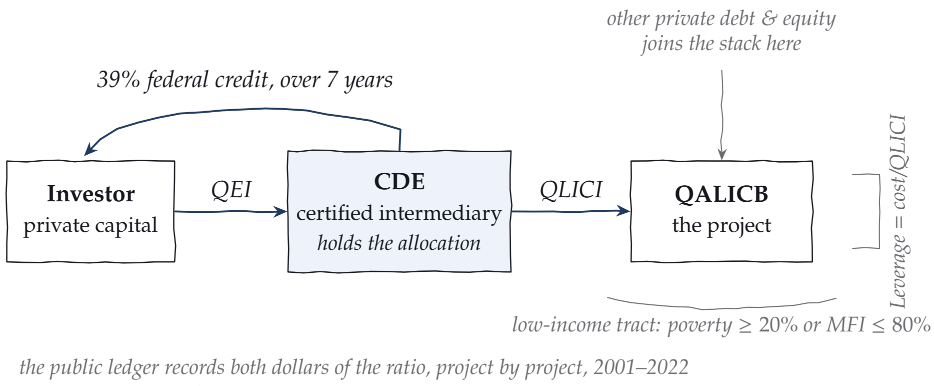

Figure 1. The program, as a working sketch. Private capital enters as a Qualified Equity Investment. The intermediary deploys it as a QLICI into a project in an eligible tract, and the 39% credit returns to the investor over seven years. The public ledger records both inputs to the leverage ratio for every project.

ing the earliest sample years outside its scope. The statute names no percentage. The familiar 20% figure comes from the CDFI Fund's administration of the requirement through allocation applications and agreements. I therefore use "administrative target" throughout.[1] Aggregate deployment is close to that line: non-metro tracts received 19.6% of QLICI dollars over the sample period.

2.3 *The intermediary layer*

The public release names 345 distinct CDEs in the transaction ledger (343 at the project level); 310 executed five or more transactions. The twenty largest account for 23.9% of QLICI dollars, so deployment is moderately concentrated. Specialization is extreme: among CDEs with at least five transactions, cumulative non-metro shares span the full range from 0% to 100%, with 39% of such CDEs never deploying non-metro and 6.5% deploying there at least four times out of five. This dispersion motivates the decomposition. An aggregate rural penalty can arise when rural deals flow through intermediaries that mobilize less *everywhere*, even if their rural and urban projects have similar leverage. The public release identifies CDEs by name. A later extension will classify their institutional form (bank subsidiary, nonprofit CDFI, for-profit specialist).

[1] The effective dates matter for the analysis; Section 5.7 reports the mandate results for the full period and for origination years from 2007 onward.

3 Data

The analysis uses the CDFI Fund's NMTC Public Data Release covering FY2003–FY2022 award years (released June 2024), which reports 19,907 QLICI transactions across 8,024 projects with origination years 2001–2022. Transactions are aggregated to the project level; the project is the unit at which total cost, and hence leverage, is defined. Cleaning steps, field definitions, and file hashes are documented in the public repository's data dictionary and provenance files, and the full pipeline from the raw workbook to every table in this paper runs from a single script sequence.

The outcome is project-level leverage,

$$\text{Leverage}_i = \frac{\text{ProjectCost}_i}{\text{QLICI}_i},$$

which I refer to as leverage throughout. Leverage is bounded below at 1 by construction, and the mass at 1 collects projects whose recorded cost equals the subsidized investment alone.

The ratio measures financing structure. The QLICI is the investment placed by the intermediary; federal cost comes through the 39% credit claimed against the qualified equity investment that funds it. That credit is spread across seven years. Under the substantially-all requirement linking the two magnitudes, a dollar of QLICI corresponds to roughly thirty-nine cents of tax expenditure. The aggregates show the difference. \$120.9 billion of project cost against \$66.6 billion of QLICI leaves \$54.3 billion of other capital, or \$0.82 for every dollar of subsidized investment and about \$2.09 for every dollar of implied tax expenditure. A constant change of denominator leaves the rural coefficient's sign, its relative magnitude, and every test below unchanged. The baseline outcome is winsorized to $[1, 20]$ to discipline a long right tail of large capital-stack deals; Section 5.4 shows the headline result is insensitive to the cap, including its removal.

Table 1 reports descriptives. Non-metro projects are 19.5% of the sample, again close to the administrative target, and lag metro projects in both mean (1.73 versus 1.99) and median (1.07 versus 1.19) leverage. Figure 3 shows the shape of the gap: non-metro mass piles up near the 1.0 floor while the metro distribution carries a fatter right shoulder.

Table 1. NMTC deployment by metro status, FY2001–FY2022.

	Metro	Non-metro
Projects	6,463	1,561
QLICI deployed ($M)	53,566	13,045
Total project cost ($M)	97,151	23,751
Mean leverage	1.99	1.73
Median leverage	1.19	1.07

4 Empirical strategy

4.1 *Layered fixed-effects decomposition*

Let R_i indicate a non-metro project. I estimate the sequence

$$L_i \;=\; \alpha + \beta R_i + \delta_{t(i)} + \eta_{q(i)} + \mu_{s(i)} + \gamma_{c(i)} + \varepsilon_i,$$

adding origination-year (δ), QALICB-type (η), state (μ), and CDE (γ) fixed effects in turn. The workhorse CDE specification substitutes intermediary effects for the state effects in the preceding column, since intermediary identity already absorbs much of the same variation. Column 7 of Table 2 carries both sets of effects and provides the strictly nested comparison. With $\gamma_{c(i)}$ included, $\hat{\beta}$ comes from rural–urban comparisons *within* an intermediary. It measures the within-CDE rural penalty, or the market-structure component. The change in $\hat{\beta}$ after adding CDE effects measures the between-CDE selection component. I cluster standard errors in these specifications at the CDE level, where both the fixed effect and the economic decision reside.

4.2 *Interpretation and limits*

This is a selection-on-observables decomposition. CDEs do not allocate deals randomly across rural and urban tracts, so I give $\hat{\beta}$ no LATE interpretation. The design establishes whether the aggregate rural penalty survives comparison within the intermediary. That comparison holds fixed its deal-structuring capacity, underwriting practice, and investor relationships. A fuzzy RDD at the LIC eligibility threshold, estimated separately for metro and non-metro samples, requires tract-level ACS covariates and remains for the follow-on paper.

The between-CDE component is an observed composition fact. Calling it "selection" adds an interpretation. Rural market conditions could

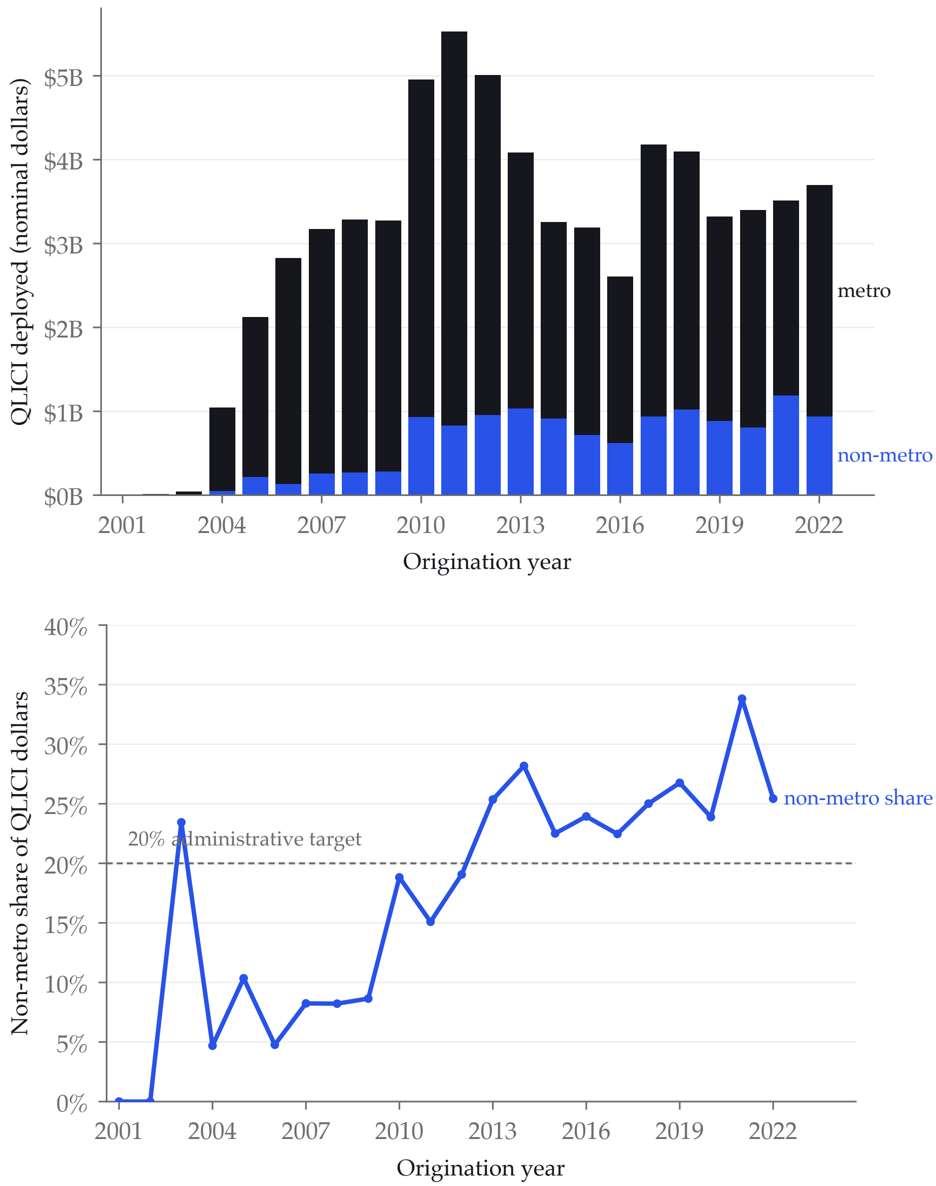

Figure 2. Deployment over time. Left: annual QLICI dollars, metro and non-metro. Right: the non-metro dollar share against the 20% administrative target.

cause low-leverage intermediaries to specialize in rural deployment, placing part of the between-CDE component in market structure operating through their business models. The within-CDE null is directly identified.

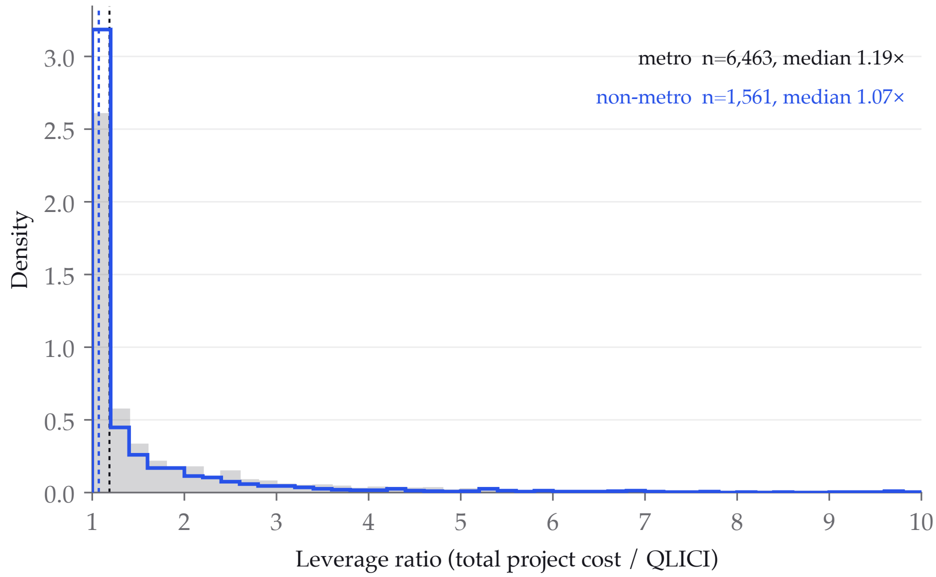

Figure 3. Project-level leverage distributions by metro status. Non-metro mass concentrates at the 1.0× floor (zero private mobilization); the metro distribution has the fatter right shoulder.

4.3 *Excess mass at the 20% target*

For each CDE j, define its cumulative non-metro deployment share s_j, computed both as a count share, $n_{j,\text{nonmetro}}/n_{j,\text{total}}$, and as a dollar share of QLICI, since the requirement is written in terms of investment. If the 20% target binds at the deployment margin, optimizing CDEs should pile up at $s_j = 0.20$ and produce excess mass in the sense of Chetty et al. [2011] and Kleven [2016]. The test compares empirical mass in a ±2.5 percentage-point window around 20% against a degree-3 polynomial counterfactual fitted outside the window, over the 310 CDEs with at least five transactions, with a CDE-level bootstrap for inference. The public data reveal no discontinuity in a CDE's objective or budget set at the threshold. Any consequence of falling short operates through future allocation rounds that the release does not observe. Because the proportionality instruction dates from 2006, I report the test for the full period and again for origination years from 2007 onward.

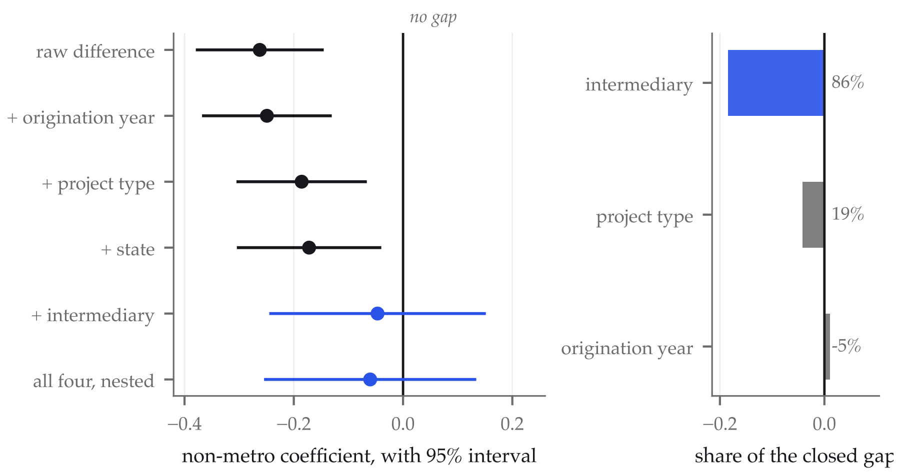

Figure 4. The rural coefficient as fixed effects enter. Left, the non-metro estimate with its 95% interval under each specification; the interval crosses zero only when intermediary identity enters, and stays there in the strictly nested model. Right, the order-invariant decomposition of the movement between the raw and full specifications. Values are read from the same pipeline outputs that produce Tables 2 and 3.

5 Results

5.1 *The decomposition*

Figure 4 puts the whole argument on one axis before the tables spell it out. Each interval is the non-metro coefficient under a successively richer specification, and the collapse happens at one step.

Table 2 reports the main estimates. The raw rural gap (column 1) is −0.262 with a heteroskedasticity-robust standard error of 0.060: non-metro projects carry about a quarter of a dollar less additional capital per dollar of subsidized investment, and the difference is precise. Origination-year effects barely move it (column 2, −0.249), leaving little scope for a vintage explanation. QALICB-type effects (column 3) shrink it to −0.186. Rural deals skew toward operating-business types with lower leverage, and this composition explains roughly a quarter of the raw gap. State effects (column 4) move it modestly further to −0.172.

Introducing CDE fixed effects in column 5 collapses the rural coefficient to −0.047 with a CDE-clustered standard error of 0.101, roughly a fifth of the raw gap in point-estimate terms. The interval, [−0.245, +0.152], is wide, and Section 5.2 quantifies how much the mean regression can and

Table 2. The rural leverage gap under layered fixed effects. Outcome: project leverage, winsorized [1, 20]. Columns (1)–(4) report HC1 standard errors; columns (5) and (7) cluster at the CDE level; column (6) reports the quantile-regression asymptotic standard error, which is not clustered. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) Median	(7) OLS
Non-metro ($\hat{\beta}$)	-0.262***	-0.249***	-0.186***	-0.172**	-0.047	-0.001	-0.060
	(0.060)	(0.061)	(0.061)	(0.067)	(0.101)	(0.008)	(0.099)
Year FE	–	✓	✓	✓	✓	✓	✓
QALICB-type FE	–	–	✓	✓	✓	✓	✓
State FE	–	–	–	✓	–	–	✓
CDE FE	–	–	–	–	✓	✓	✓
R^2	0.002	0.015	0.022	0.039	0.131	–	0.141
N	8,024	8,024	8,024	8,024	8,024	8,024	8,024

cannot rule out. The median specification (column 6) is precise, placing the within-CDE rural penalty at −0.001 (SE 0.008). Because column 5 substitutes CDE effects for the state effects of column 4, column 7 reports the strictly nested model carrying year, type, state, and CDE effects together; it returns −0.060 (SE 0.099, $p = 0.54$), so no part of the reading depends on that substitution.

Because sequential addition of fixed-effect blocks is order-dependent, Table 3 reports the Gelbach [2016] decomposition with the full model as reference, which is invariant to listing order. Of the −0.216 movement from the raw to the full specification, CDE identity contributes −0.185 (86% of the movement; CDE-cluster bootstrap 95% CI [−0.347, −0.023]), QALICB type −0.041, and origination year a small offsetting +0.010. The share version of this statement, 82% of the raw gap, carries a wide bootstrap interval ([25%, 255%]) because it is a ratio of estimates. The level estimate is more informative. The CDE composition component is large, negative, and statistically distinct from zero. The within-CDE component is statistically indistinguishable from zero.

5.2 *Precision, power, and margins*

The mean and median estimates differ in precision by an order of magnitude. For the mean specification, a one-sided 5% test can reject within-CDE rural penalties larger than 0.213, nearly the size of the raw gap. The mean regression alone therefore gives weak evidence for a zero; the outcome's

Table 3. Gelbach (order-invariant) decomposition of the raw rural gap, full-model reference. Contributions sum to $\hat{\beta}_{M0} - \hat{\beta}_{\text{full}}$ by identity (verified to 10^{-14} in the replication code).

	Contribution	Share of movement
Origination year	+0.010	-4.8%
QALICB type	-0.041	19.2%
CDE identity	-0.184	85.6%
Total ($\hat{\beta}_{M0} - \hat{\beta}_{\text{full}}$)	-0.216	100%
CDE component bootstrap 95% CI: [-0.347, -0.023] (499 CDE-cluster reps)		

long right tail, which inflates mean-regression variance, is the proximate reason. The median specification can reject penalties larger than 0.016. The median estimate and the margins carry the null; I report the mean for comparability.

Because the median estimate carries so much of the argument, I subject its precision to three further tests, reported in Table 4. A CDE-cluster pairs bootstrap over 500 replications, refitting the full median regression on each draw and treating an intermediary drawn twice as two intermediaries with their own fixed effects, returns a standard error of 0.0093 against the asymptotic 0.0076. Clustering therefore inflates the median standard error by about a quarter, which moves the rejectable penalty from 0.013 to 0.016 and leaves the substantive claim intact. That inflation is specific to the resampling scheme, and I report the conservative one deliberately. An exchangeably weighted cluster-multiplier bootstrap, drawing independent unit-mean weights at the intermediary level, returns 0.0075, which is essentially the asymptotic value and would leave the bound at 0.013. Two defensible cluster bootstraps disagreeing by a quarter is itself a symptom of the nonregularity described below, and I take the wider of the two throughout. Randomization inference, permuting the rural label inside each intermediary, returns a two-sided p of 0.085 against the sharp null of no within-CDE rural effect. That figure carries an assumption worth naming. Permuting within intermediary holds each intermediary's rural count fixed and nothing else, while rural status is strongly associated with both origination year and project type, so the permutation destroys structure the specification conditions on. Under the within-CDE permutation null, the observed project-type composition sits 9.2 standard deviations from its own reference distribution. I therefore treat 0.085 as a sensitivity that presumes exchangeability conditional on intermediary alone, and I report the stricter test beside it. Permuting only inside exact intermediary-by-year-by-type strata holds every conditioned margin fixed by construction and returns 0.170, resting on the 460 of 3,255 exact strata that are mixed in rural status and cover 2,688 projects. Neither test rejects, and the within-CDE figure itself moves between 0.085 and 0.100 across independent permutation streams, which is a further reason to lean on the interval instead of on either p-value.

One caution belongs with these numbers. The outcome carries a 26.9% point mass at exactly 1.0, the value recorded when a project mobilizes nothing beyond the subsidized investment, and the unconditional median of 1.159 sits on the shoulder of that mass. Quantile-regression asymptotic standard errors are sparsity estimates that presume a positive continuous conditional density at the estimated quantile, and this configuration violates that presumption. The symptom is visible in the randomization distribution, where 91% of permutations return a rural coefficient pinned to within 10^{-7} of zero instead of varying smoothly, which is the signature of a degenerate vertex solution. I therefore report the bootstrap standard error, which makes no density assumption, and I note that it and the asymptotic one happen to agree closely here. Readers should treat the median specification as well identified in its location and poorly identified in its curvature.

Figure 5 carries this further by estimating the same specification across the quantile process. The coefficient sits at zero through the lower half of the distribution, then drifts negative above the median, reaching −0.129 at the 0.90 quantile and −0.200 at 0.95. Read alone, that gradient suggests a rural penalty concentrated among the deals that mobilize the most private capital. The bootstrap does not sustain the reading. Intervals widen at least as fast as the point estimates fall, giving cluster-bootstrap standard errors of 0.125 and 0.215 at those two quantiles, so no point in the upper tail is distinguishable from zero. I report the gradient because its monotonicity is suggestive and because a design with more power in the upper tail is a natural next test, and I decline to interpret it as a result.

Figure 6 shows the identifying variation. Among the 104 dual-market intermediaries with at least three projects on each side of the rural line, each vertical spine connects a CDE's urban-book mean leverage to its rural-book mean. The spines climb a staircase from 1.0× to 4.0× across intermediaries, while the within-spine gaps distribute tightly around zero (median −0.12, interquartile range [−0.46, +0.36]). The range of levels across intermediaries is several times wider than the spread of gaps within them.

Table 4. Inference for the median specification. The bootstrap is a CDE-cluster pairs bootstrap over 500 replications; randomization inference permutes the rural label within each intermediary over 400 draws. The paired test uses no regression, comparing each dual-market intermediary's rural median with its own urban median.

	Value
Point estimate, within-CDE rural coefficient	−0.0006
Asymptotic standard error (quantile regression)	0.0076
CDE-cluster bootstrap standard error (500 reps)	0.0093
Inflation from clustering	1.23×
Bootstrap percentile 95% interval	[−0.032, −0.000]
Rejectable penalty, asymptotic	0.013
Rejectable penalty, clustered	0.016
Randomization inference, two-sided p (400 draws)	0.085
Permutations pinned to a vertex	91%
Paired within-CDE median gap (104 intermediaries)	−0.018
Wilcoxon signed-rank p on the paired gaps	0.140

The margins address a second way a mean could mislead: measured at a threshold of 1.001, 27.4% of projects record no capital beyond the subsidized investment, so a within-CDE penalty could in principle hide in the probability of drawing any additional capital at all. It does not. Within CDE, rural deals are 1.8 percentage points less likely to clear the floor (SE 1.8pp, $p = 0.34$), and among deals that mobilize, the rural coefficient is −0.039 ($p = 0.77$). Both margins are stable when the floor is moved to 1.05. Identification for all within-CDE results comes from the 163 of 343 CDEs that deploy on both sides of the rural line; these switchers originate 94% of all rural projects.

5.3 *Heterogeneity by project type*

Interacting the rural indicator with QALICB type (with CDE and year fixed effects retained) yields uniformly imprecise interactions: the rural main effect is 0.091 (SE 0.621), and no type-specific rural penalty is distinguishable from zero. No single project type concentrates or masks the within-CDE null. The point estimates weakly echo the descriptive pattern, with the most negative estimate for real estate, where private debt-stacking is most natural. The data cannot reject homogeneity.

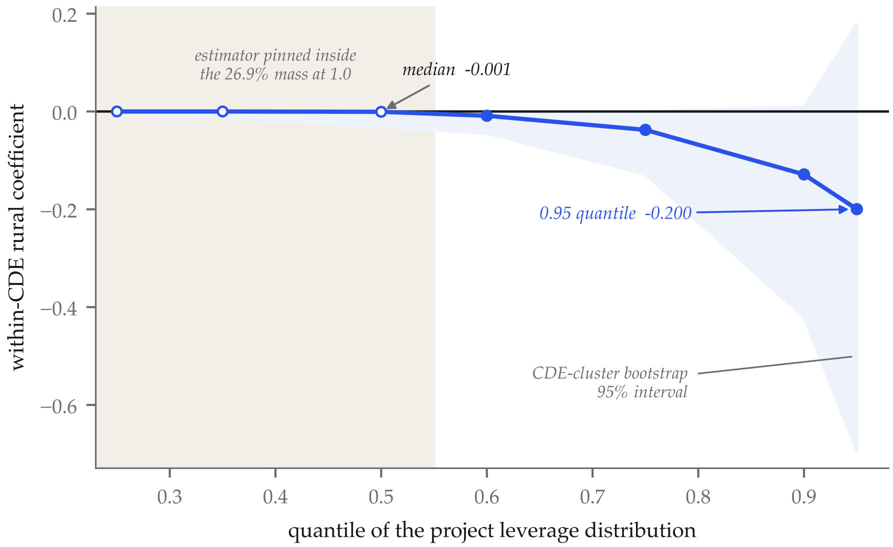

Figure 5. The within-CDE rural coefficient across the quantile process, with CDE-cluster bootstrap 95% intervals. Open markers and the shaded region mark quantiles where the estimator is pinned inside the 26.9% point mass at 1.0 and its standard error is not interpretable; the boundary is read from the share of bootstrap draws stuck at the point estimate rather than assumed. The point estimates drift negative above the median, and the intervals widen at least as quickly, so the upper tail is not distinguishable from zero.

5.4 *Robustness*

Table 5 varies the outcome transformation, winsorization cap, sample period, project sample, and clustering scheme. Unwinsorizing the outcome makes the naive gap *larger* (−0.338, $p = 0.01$), while the within-CDE estimate remains null (−0.216, $p = 0.48$). Winsorization does not produce the main result. A log outcome puts the within-CDE penalty at −3.2% ($p = 0.18$). Tightening or loosening the winsorization cap ($[1, 10]$, $[1, 50]$), splitting the sample at 2010, and restricting to single-CDE projects all leave the within-CDE rural coefficient statistically indistinguishable from zero. Restricting to the fifty largest CDEs, where the fixed effects are estimated most precisely, gives −0.100 (SE 0.148); two-way clustering by CDE and census tract moves the standard error from 0.101 to 0.105.

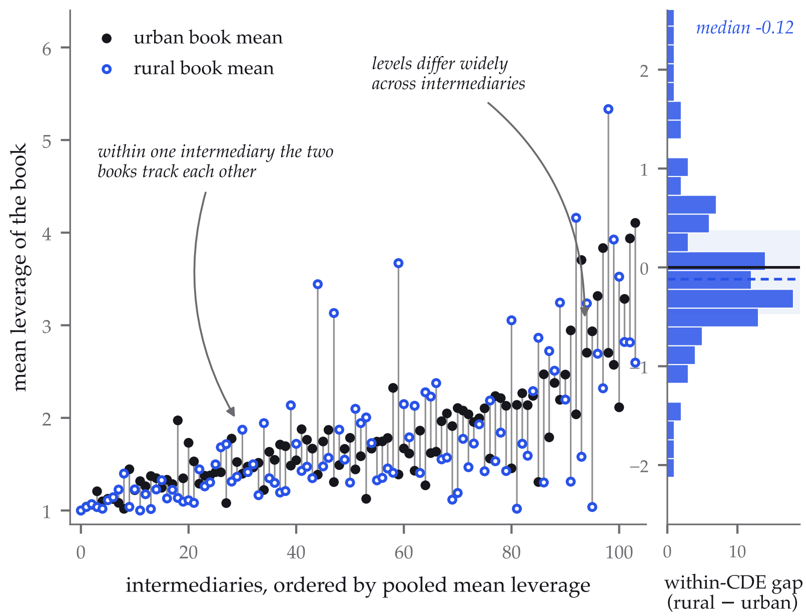

Figure 6. The switchboard. Every intermediary with at least three urban and three rural projects appears as a spine from its urban-book mean (filled) to its rural-book mean (open), ordered by pooled mean leverage; the marginal panel gives the distribution of within-CDE gaps against zero. The figure's sample rule and every displayed statistic are recorded in a machine-readable sidecar alongside the figure in the repository.

5.5 *Purpose of investment*

The public release records a purpose of investment on every transaction, and the project-level file this analysis is built from does not carry that field, so none of the specifications above use it. The omission deserves attention because purpose is a composition channel of exactly the kind the decomposition exists to separate from intermediary identity. I construct a project-level purpose by taking the category accounting for the largest share of a project's QLICI dollars, which covers every project in the estimation sample; 7.4% of projects combine more than one purpose, and the median dominant share is 100%.

The raw association runs in the direction that would concern a referee. Business financing accounts for 59.5% of rural projects against 34.8% of metro projects, and it carries the lowest median leverage of any category at 1.044. Commercial real estate rehabilitation, with a median of 1.271, accounts for 28.1% of metro projects and 12.8% of rural ones. Rural deployment is weighted toward the purpose that mobilizes least, which raises the

Table 5. Robustness of the within-CDE rural coefficient. All rows except the first include year, QALICB-type, and CDE fixed effects with CDE-clustered standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	$\hat{\beta}$	(SE)	p	N
Unwinsorized outcome, no FE	-0.338**	(0.136)	0.01	8,024
Unwinsorized outcome	-0.215	(0.304)	0.48	8,024
Log outcome	-0.032	(0.024)	0.18	8,024
Winsorized at [1, 10]	-0.066	(0.073)	0.37	8,024
Winsorized at [1, 50]	-0.047	(0.138)	0.73	8,024
Origination 2001–2009	-0.044	(0.147)	0.77	2,488
Origination 2010–2022	-0.039	(0.107)	0.71	5,536
Single-CDE projects only	-0.088	(0.139)	0.53	6,366
Top-50 CDEs only	-0.100	(0.148)	0.50	4,229
Two-way cluster (CDE × tract)	-0.047	(0.105)	0.66	8,024
Bunching excess mass $\hat{B}$	-0.0006	95% CI [-0.0136, 0.0135], 999 CDE-bootstrap reps		

possibility that some of the raw gap reflects what the money does instead of who deploys it.

Table 6 shows that the possibility does not survive measurement. Purpose entered alongside year effects alone moves the rural coefficient to −0.2279, a smaller movement than QALICB type produces on its own. Added to a specification already carrying type, purpose moves the coefficient the other way, to −0.1946. In the order-invariant decomposition with purpose as a fourth block, its contribution is +0.0153, which offsets rather than explains, and the CDE contribution rises to −0.1824, or 88.1% of the explained movement against the 86% of Table 3. The within-CDE coefficient moves from −0.0467 to −0.0551 and remains indistinguishable from zero ($p = 0.578$).

Two readings are available and I prefer the conservative one. Purpose and QALICB type are strongly related, since a real estate borrower largely does real estate, so the two blocks compete for the same variation and the split between them should not be over-interpreted. What the exercise establishes is narrower and sufficient: the headline is not an artifact of the missing field. Allowing purpose to compete for the explained movement leaves intermediary identity carrying more of it, not less.

5.6 *Where the residual lives*

An average null can conceal a subpopulation, so the workhorse specification was re-estimated inside twenty-four subgroups: quartiles of interme-

Table 6. Purpose of investment, a channel recorded in the public release and not used elsewhere in this paper. Panel A gives composition and leverage by purpose. Panel B inserts purpose into the ladder; columns through the sixth row report HC1 standard errors and the last two cluster at the CDE level. Panel C reports the order-invariant decomposition with purpose as a fourth block, whose contributions sum to the total by identity (residual below 10^{-15}).

Panel A. Composition and leverage by purpose

Purpose	Projects	Median lev.	% rural
Business financing	3,177	1.044	59.5%
Real estate, construction	2,432	1.208	24.6%
Real estate, rehabilitation	2,015	1.271	12.8%
Housing, single family, new	163	1.601	0.7%
Other financing purpose	125	1.000	2.4%
Folded rare categories	112	1.037	0.1%

Panel B. The rural coefficient with purpose inserted

Specification	$\hat{\beta}_{\text{rural}}$	(SE)
Raw	−0.2623	(0.0597)
Year	−0.2492	(0.0606)
Year, QALICB type	−0.1856	(0.0609)
Year, purpose	−0.2279	(0.0620)
Year, type, purpose	−0.1946	(0.0617)
Year, type, purpose, state	−0.1765	(0.0679)
Year, type, CDE	−0.0467	(0.1011)
Year, type, purpose, CDE	−0.0551	(0.0990)

Panel C. Order-invariant decomposition, purpose as a fourth block

Block	Contribution	Share
Origination year	+0.0110	−5.3%
QALICB type	−0.0510	24.6%
Purpose of investment	+0.0153	−7.4%
CDE identity	−0.1824	88.1%
Total explained	−0.2071	

diary size, three origination eras, the four census regions, the four project types, three bands of the intermediary's own rural orientation, and six categories of purpose of investment. Four cells hold too few deals to identify a within-CDE comparison and are reported rather than dropped. Table 8 gives the other twenty.

Three of the twenty estimated cells reach conventional significance without correction. Two of them behave as a scan of this size should: real-estate projects at −0.395 ($p = 0.043$) and the third quartile of intermediary size at +0.349 ($p = 0.048$) point in opposite directions, and across twenty tests at the 5% level chance alone produces roughly one such rejection. Neither survives a Benjamini-Hochberg correction.

The third does not behave that way. Within intermediary, rural commercial real-estate rehabilitation projects carry a rural coefficient of −0.4393 with a CDE-clustered standard error of 0.1019, a t statistic of −4.31, and a p-value of 1.6×10^{-5}. It is the only cell in the scan to survive the Benjamini-Hochberg correction, and it is separated from the rest of the scan by more than three orders of magnitude in p. The estimate is large as well as precise: relative to the metro rehabilitation deals originated by the same intermediary in the same year, rural rehabilitation mobilizes roughly forty-four cents less per subsidized dollar, or about 15% in logs.

Because a result of this kind invites the suspicion that it was found by looking, I state its provenance plainly. The purpose cells were added to this scan after the QALICB real-estate cell had been flagged in an earlier draft as the one worth a sharper test, and the Benjamini-Hochberg correction covers the twenty cells in the scan without covering the decision to add the purpose dimension. No within-scan correction repairs that, so the finding is exploratory by construction. What can be done is to attack it, and Table 7 reports six attempts.

The estimate does not depend on the outcome transformation: it lies between −0.403 and −0.463 across the unwinsorized outcome and three winsorization caps, and the log specification gives −0.155 ($p = 0.0002$). It does not depend on any one intermediary or state. Dropping each of the 256 intermediaries in turn moves the coefficient only within $[-0.481, -0.410]$, and dropping each state in turn keeps it within $[-0.478, -0.380]$, with the sign never changing in any of those regressions. Randomization inference, permuting the rural label inside each intermediary over 2,000 draws, returns $p = 0.0055$, and a wild cluster bootstrap-t over 2,000 draws returns $p = 0.0005$. Running the identical pipeline on the adjacent purpose categories finds nothing: business financing gives +0.053 ($p = 0.74$) and real-estate construction gives −0.068 ($p = 0.63$).

Two further checks locate the estimate and qualify it. Rebuilding the cell from the raw workbook with an independent code path returns the same −0.4393 and 0.1019, and the result does not depend on how purpose is assigned: taking each project's modal purpose by transaction count gives −0.4570, and restricting to projects whose transactions all share a single purpose, which removes the assignment rule entirely, gives −0.4014. Restricting to single-CDE projects, where intermediary attribution is unambiguous, gives −0.4973. The qualification concerns where inside the cell the effect sits. Quantile regressions within the cell put the rural coefficient at −0.010 at the lower quartile and −0.890 at the ninetieth percentile, and trimming the outer decile of the leverage distribution moves the mean estimate to −0.2730 (SE 0.0820, $p = 0.0009$). The effect is therefore real away from the tails and substantially larger within them, so the headline figure should be read as a mean over a distribution whose upper reaches carry 38% of it.

That last contrast is the most informative of the six. Construction and rehabilitation are both commercial real estate and both are classified under the same QALICB type, which is why no specification in this paper could previously separate them. They diverge sharply here, and the divergence runs in the direction the mechanism predicts, since rehabilitation carries the historic-credit and layered-debt structures in which additional private capital is most readily stacked and rural markets are least able to supply it. I therefore read the paper's headline as holding on average and as admitting one specific, robust, and economically interpretable exception. Confirming it on an independent sample is the natural next step, and until that exists the result should be treated as a well-tested hypothesis instead of an established fact.

5.7 No excess mass at the 20% target

Figure 7 plots the cross-CDE distribution of cumulative non-metro shares against the polynomial counterfactual. Excess mass in the window around the 20% administrative target is $\hat{B} = -0.0006$, which is *negative* and −2.2% of the counterfactual mass, with a CDE-level bootstrap 95% confidence interval of [−0.014, +0.013]; 47% of bootstrap draws are positive. There is no evidence that CDEs manage their cumulative deployment to the mandate line.

I repeat the test using QLICI dollars, which matches how the requirement is written, gives excess mass of +0.005 with a bootstrap interval of [−0.008, +0.020]. Restricting to origination years from 2007 onward, after

Table 7. Attempts to break the commercial real-estate rehabilitation cell. The baseline is the workhorse specification estimated inside that cell, with year, QALICB-type and CDE fixed effects and CDE-clustered standard errors. Randomization inference permutes rural within intermediary; the wild cluster bootstrap-t imposes the null and uses Rademacher weights.

Check	Specification or statistic	Result
Baseline	$\hat{\beta}_{\text{rural}}$ (CDE-clustered SE)	−0.4393 (0.1019)
	t statistic	−4.31
	p-value	1.6×10^{-5}
Outcome	Winsorized [1, 20] (baseline)	−0.4393 (0.1019)
	Unwinsorized	−0.4634 (0.1041)
	Log outcome	−0.1548 (0.0417)
	Winsorized [1, 10]	−0.4031 (0.0956)
	Winsorized [1, 50]	−0.4507 (0.1047)
Influence	Leave one CDE out (256 drops)	[−0.481, −0.410]
	Leave one state out	[−0.478, −0.380]
Inference	Randomization, 2,000 draws	$p = 0.0055$
	Wild cluster bootstrap-t, 2,000 draws	$p = 0.0005$
Placebo purposes	Business financing	+0.0527 (0.1606)
	Real estate, construction	−0.0675 (0.1397)

the proportionality instruction enters the code, gives −0.003 in counts and +0.001 in dollars, with intervals of [−0.016, +0.011] and [−0.013, +0.016] across 291 intermediaries.

The mandate may bind at the allocation-award level through commitments in CDE allocation agreements. The public release does not report those commitments, and realized cumulative deployment could fall elsewhere. The cross-CDE distribution around 20% also is close to bimodal. Urban specialists sit near zero, while rural specialists sit far above 20%, leaving the mandate line in a low-density valley.

6 Discussion

Table 8. The within-CDE rural coefficient inside subgroups. Each cell repeats the workhorse specification with year, QALICB-type and CDE fixed effects and CDE-clustered standard errors. The final column reports whether the cell survives a Benjamini-Hochberg correction at the 5% level across all twenty estimated cells.

	$\hat{\beta}$	(SE)	p	N	switchers	BH 5%
CDE size quartile 1 (0<n≤4)	-0.838	(0.761)	0.27	203	10	no
CDE size quartile 2 (4<n≤13)	-0.179	(0.198)	0.36	670	32	no
CDE size quartile 3 (13<n≤30)	0.349	(0.177)	0.05	1,721	53	no
CDE size quartile 4 (30<n≤∞)	-0.130	(0.122)	0.29	5,430	68	no
Origination 2001-2009	-0.044	(0.147)	0.77	2,488	57	no
Origination 2010-2015	0.102	(0.173)	0.56	2,506	103	no
Origination 2016-2022	-0.191	(0.128)	0.14	3,030	107	no
Region Northeast	0.090	(0.261)	0.73	1,189	26	no
Region Midwest	-0.162	(0.150)	0.28	2,827	60	no
Region South	0.105	(0.141)	0.46	2,432	103	no
Region West	-0.322	(0.214)	0.13	1,538	48	no
QALICB type CDE	*not estimated*			91	4	
QALICB type NRE	0.043	(0.153)	0.78	3,775	123	no
QALICB type RE	-0.395	(0.195)	0.04	2,862	87	no
QALICB type SPE	-0.075	(0.142)	0.60	1,296	66	no
CDE rural share ≤20%	-0.066	(0.157)	0.68	3,481	76	no
CDE rural share 20-50%	-0.023	(0.196)	0.91	1,651	57	no
CDE rural share >50%	-0.132	(0.118)	0.26	969	30	no
Purpose: Business financing	0.053	(0.161)	0.74	3,177	106	no
Purpose: Housing, single family, new	*not estimated*			163	5	
Purpose: Other financing purpose	*not estimated*			125	5	
Purpose: Folded rare categories	*not estimated*			112	0	
Purpose: Real estate, construction	-0.068	(0.140)	0.63	2,432	94	no
Purpose: Real estate, rehabilitation	-0.439	(0.102)	<0.001	2,015	80	**yes**

6.1 *Scope of the results*

The within-CDE rural penalty is precisely zero at the median. Both margins are null, and the mean estimate is small but imprecise. The real aggregate rural penalty lies in between-CDE composition. This does not make rural and urban credit markets equivalent. It means that NMTC deployment data detect no difference once the intermediary is held fixed.

The mechanism behind composition remains open. Intermediary selection is a natural reading of the between-CDE component. Rural market conditions could also lead low-leverage intermediaries to specialize in rural deployment, placing part of that component on the market-structure side of the ledger. Separating these accounts requires variation in CDE–market assignment that the public release lacks. The estimates also aver-

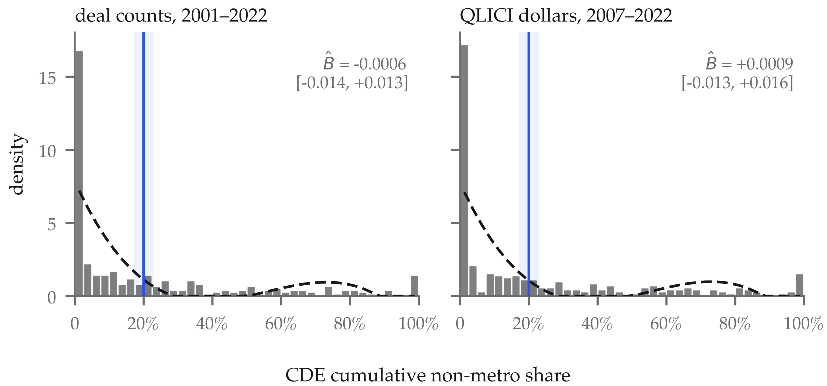

Figure 7. No excess mass at the 20% target. Cross-CDE distributions of cumulative non-metro share, over CDEs with at least five transactions, with a degree-3 polynomial counterfactual fitted outside the shaded ±2.5 percentage-point window. Left, deal counts across the full period (n = 310). Right, QLICI dollars from 2007 onward (n = 291), which matches both the units the requirement is written in and the years it governs. Each panel reports its excess mass with a CDE-cluster bootstrap interval; both contain zero. The distribution is close to bimodal, leaving the target in a low-density valley between urban and rural specialists.

age over the fixed-effects distribution; they do not set a ceiling on what any particular CDE could mobilize in rural tracts.

6.2 Policy reading

Under the existing program design, the marginal rural deal does not appear constrained by market structure: a given intermediary's mobilization does not measurably fall when it goes rural. The aggregate gap is therefore amenable to *intermediary-allocation* levers. Allocations could shift toward CDEs with demonstrated high-leverage capacity, independent of their current rural orientation, without redesigning the credit itself.

The 20% mandate operates on cumulative deployment share, where the data show no bunching. If the policy goal is rural mobilization instead of rural deal counts, that is the wrong margin.

6.3 Research program

The next step is a multi-program U.S. comparison covering NMTC, LIHTC, and Opportunity Zones, followed by an international extension to development-bank project finance. The same mobilization framework applies in each setting. More immediately, the LIC-eligibility regression discontinuity requires the ACS tract merge.

7 Conclusion

Directly observed project-level leverage places the NMTC's aggregate rural mobilization penalty between intermediaries. The order-invariant decomposition assigns −0.185 of the raw −0.262 gap to CDE identity. The within-CDE penalty is precisely zero at the median and null at both margins. Its mean estimate is imprecise and roughly a fifth of the raw gap; a strictly nested model with state and intermediary effects returns −0.060. Scanning twenty-four subgroups by intermediary size, era, region, project type, rural orientation and purpose of investment leaves one cell standing after a correction for scanning: rural commercial real-estate rehabilitation, at −0.439, which survives every transformation, drop, placebo and resampling test I put to it. That dimension was added after an earlier draft flagged real estate, so I hold it as exploratory. The cross-CDE distribution of non-metro shares shows no bunching at the 20% administrative target. Intermediary composition may itself respond to rural market conditions, so its mechanism remains unsettled. For this program, the evidence points to allocation levers operating through the CDE layer.

Contributors and authorship

Ian Helfrich is the sole author and originated the study. Katia Antunes and Elizaveta Gonchar contributed to the project. Contributor roles are stated at this level of generality because no more specific allocation has been jointly verified. Responsibility for the analysis, interpretation, and manuscript rests with the author.

A Reproducibility

The public repository regenerates every number in this draft, and a machine check enforces that claim rather than asserting it.

Building the panel. `scripts/describe_nmtc.py` builds the analysis files from the raw CDFI workbook, with hashes recorded in `PROVENANCE.md`. `scripts/run_purpose_channel.py` derives the project-level purpose of investment from the transaction sheet, which the project sheet does not carry.

Estimation. `scripts/run_regressions.py` produces Table 2 and the bunching statistics; `scripts/run_robustness.py` produces Table 5; `scripts/run_referee_fixes.py` and `scripts/run_review_round2.py` produce the decomposition, power and mandate quantities; `scripts/run_residual_analysis.py` runs the subgroup scan and its multiplicity correction; `scripts/run_median_inference.py` produces the median bootstrap, the randomization test and the quantile process; and `scripts/run_conditioned_randomization.py` produces the covariate-conditioned permutation test.

Quantile regression. `scripts/qreg_lp.py` solves the quantile regressions as sparse linear programs. With 366 of 368 columns being indicators, the dense iteratively reweighted least squares used by standard packages is both slower and less accurate here; the linear program attains a no-worse check-loss objective at every quantile tested.

Adversarial checks. `scripts/verify_value_added.py`, `scripts/verify_quantile_tail.py`, `scripts/verify_rehab_cell.py` and `scripts/validate_rehab_finding.py` attempt to break the results rather than confirm them. The last of these rebuilds the rehabilitation cell from the raw workbook along an independent code path, so that a defect in the processed pipeline would appear as a disagreement.

Exhibits. `scripts/make_paper_tables.py`, `make_residual_table.py`, `make_purpose_table.py`, `make_rehab_table.py` and `make_median_inference_table.py` write the table fragments; `scripts/make_paper_figures.py`, `make_paper_art.py` and `make_quantile_figure.py` render the figures. No table or figure in this paper is maintained by hand.

The numbers audit. `scripts/audit_paper_numbers.py` pins every load-bearing number in the manuscript to the pipeline field that produced it, matching the literal string as typeset and requiring that the source value, rounded half-up to the precision printed here, equal the printed number exactly. The audit currently covers 204 claims. It gates the Overleaf bundle, so a number that has drifted from its source cannot reach a submitted draft. `scripts/run_tests.sh` runs the test suite, including tests that confirm the audit fails when a value is perturbed. The repository also serves a browser-based viewer with the full project map and specification explorer.

References

Martin D. Abravanel, Nancy Pindus, Brett Theodos, Kassie Bertumen, Rachel Brash, and Zach McDade. New markets tax credit program evaluation: Final report. Technical report, Urban Institute, for the CDFI Fund, 2013.

Samantha Attridge and Lars Engen. Blended finance in the poorest countries: The need for a better approach. Technical report, Overseas Development Institute, 2019.

Raj Chetty, John N. Friedman, Tøre Olsen, and Luigi Pistaferri. Adjustment costs, firm responses, and micro vs. macro labor supply elasticities: Evidence from Danish tax records. *Quarterly Journal of Economics*, 126(2): 749–804, 2011.

Convergence Finance. State of blended finance 2024. Technical report, Convergence, 2024.

Kevin Corinth, David Coyne, Naomi Feldman, and Craig E. Johnson. The targeting of place-based policies: The New Markets Tax Credit versus Opportunity Zones, 2024. NBER chapter; SSRN 5122131.

Rebecca Diamond and Tim McQuade. Who wants affordable housing in their backyard? an equilibrium analysis of low-income property development. *Journal of Political Economy*, 127(3):1063–1117, 2019.

Matthew Freedman. Teaching new markets old tricks: The effects of subsidized investment on low-income neighborhoods. *Journal of Public Economics*, 96(11–12):1000–1014, 2012.

Matthew Freedman and Annemarie Kuhns. Supply-side subsidies to improve food access and dietary outcomes: Evidence from the New Markets Tax Credit. *Urban Studies*, 55(14):3234–3252, 2018.

Jonah B. Gelbach. When do covariates matter? and which ones, and how much? *Journal of Labor Economics*, 34(2):509–543, 2016.

Edward L. Glaeser and Joshua D. Gottlieb. The economics of place-making policies. *Brookings Papers on Economic Activity*, 2008(1):155–253, 2008.

Tami Gurley-Calvez, Thomas J. Gilbert, Katherine Harper, Donald J. Marples, and Kevin Daly. Do tax incentives affect investment? an analysis of the New Markets Tax Credit. *Public Finance Review*, 37(4):371–398, 2009.

Kaitlyn Harger and Amanda Ross. Do capital tax incentives attract new businesses? evidence across industries from the New Markets Tax Credit. *Journal of Regional Science*, 56(5):733–753, 2016.

Henrik Jacobsen Kleven. Bunching. *Annual Review of Economics*, 8:435–464, 2016.

Patrick Kline and Enrico Moretti. People, places, and public policy: Some simple welfare economics of local economic development programs. *Annual Review of Economics*, 6:629–662, 2014.

Hans Peter Lankes. Blended finance for scaling up climate and nature investments. Technical report, CEPR / LSE, 2021.

David Neumark and Helen Simpson. Place-based policies. In Gilles Duranton, J. Vernon Henderson, and William C. Strange, editors, *Handbook of Regional and Urban Economics*, volume 5B, chapter 18. Elsevier, 2015.

Brett Theodos, Ananya Hariharan, Jorge González-Hermoso, and Sophie Edelman. Where do new markets tax credit projects go? Technical report, Urban Institute, 2020.

Brett Theodos, Christina Plerhoples Stacy, Daniel Teles, Christopher Davis, Prasanna Rajasekaran, and Ananya Hariharan. Which community development entities receive NMTC funding? Technical report, Urban Institute, 2021.

Brett Theodos, Christina Stacy, Daniel Teles, Christopher Davis, and Ananya Hariharan. Place-based investment and neighborhood change: The impacts of the New Markets Tax Credit on jobs, poverty, and demographic composition. *Journal of Regional Science*, 62(4):1092–1121, 2022.